

Supplementary Material

Segmentation Error and Boundary-Condition Tuning in Computed Coronary FFR: A Controlled In Silico Study

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S1. ANALYSIS PIPELINE

Fig. S1 shows the analysis pipeline. Host vessels were eligible with an image quality of at least 2 (0–4 scale), a taper-fit radius r_{fit} of at least 1.0 mm at the lesion center and native narrowing below 40% DS. The measurement point 20 mm beyond the lesion lay on a vessel of radius at least 0.75 mm, and the FFR before insertion was at least 0.90.

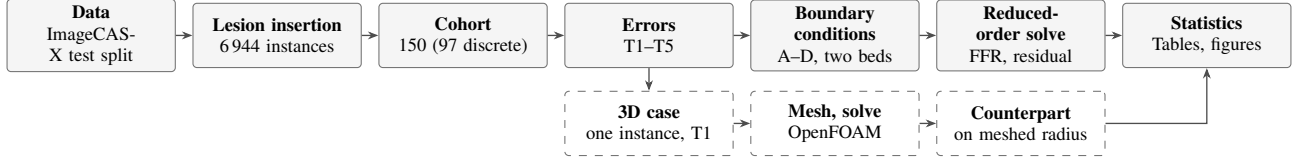


Fig. S1. Analysis pipeline. Shaded boxes: reduced-order arm, run for every instance in one automated run. Dashed boxes: the 3D case study. Sensitivity analyses and noise floors repeat the reduced-order arm.

S2. EXCLUSIONS

Table S1 lists every exclusion. Protocols C and D require at least two territories that retain a vessel. A Protocol C fit at its search bound is a failure (two half-voxel throat-error models; 10 of 4940 noise-floor draws). Protocol D fits at the $10^{\pm 3}$ bound keep a defined residual and are retained (two missed-branch models, residual 0.08 and 0.11; 23 half-voxel and 41 ten-point throat-error models with an increased stenosis, of which 43 failed the check and 21 passed while materially wrong). Under Protocol A a vessel break can remove every node with bed outflow (36 discrete, 2 leaky); scored as flips (FFR 1), these raise the discrete-bed vessel-break flip rate from 40% to 43%.

Table S1. Cells With Excluded Models

Bed	Error	Protocol	Solved	Excluded (reason: count)
Discrete	T1	all	77 of 97	no side branch distal to the lesion: 20
Discrete	T1	C	44	fewer than two territories with surviving vessels: 33
Discrete	T2	all	96 of 97	truncation point at a segment start: 1
Discrete	T2	A	60	no node with bed outflow remains: 36
Discrete	T2	C	60	fewer than two territories with surviving vessels: 36
Discrete	T5	C	193	Protocol C fit at search bound: 1
Leaky	T1	all	118 of 150	no side branch distal to the lesion: 32
Leaky	T1	C	71	fewer than two territories with surviving vessels: 47
Leaky	T2	all	149 of 150	truncation point at a segment start: 1
Leaky	T2	A	147	no node with bed outflow remains: 2
Leaky	T2	C	100	fewer than two territories with surviving vessels: 49
Leaky	T5	C	299	Protocol C fit at search bound: 1

S3. MIXED-EFFECTS MODEL OF FLIPS

Table S2. Bayesian Mixed-Effects Logistic Model of Flips: Posterior Mean Log-Odds (95% Interval)

Term	Discrete bed	Leaky bed
Intercept	-2.63 (-2.85, -2.42)	-2.39 (-2.61, -2.18)
T2	0.69 (0.31, 1.07)	1.81 (1.47, 2.15)
T3	-1.92 (-2.40, -1.44)	-2.49 (-3.12, -1.87)
T4	-1.16 (-1.59, -0.73)	-1.00 (-1.45, -0.54)
B	-1.54 (-1.92, -1.17)	-1.76 (-2.27, -1.26)
C	-0.69 (-1.10, -0.28)	-0.99 (-1.42, -0.57)
T2 × B	0.06 (-0.55, 0.66)	-1.89 (-2.66, -1.12)
T3 × B	1.26 (0.41, 2.10)	1.13 (-0.06, 2.32)
T4 × B	0.72 (-0.08, 1.51)	-1.26 (-2.87, 0.35)
T2 × C	-0.73 (-1.46, 0.00)	-2.00 (-2.76, -1.24)
T3 × C	0.44 (-0.39, 1.28)	0.78 (-0.25, 1.81)
T4 × C	-0.10 (-0.89, 0.68)	0.21 (-0.54, 0.96)
Severity (% DS)	0.019 (0.016, 0.022)	0.019 (0.016, 0.022)
Length (mm)	-0.04 (-0.05, -0.02)	-0.08 (-0.09, -0.06)

Reference levels: T1 (missed branch) and Protocol A. Baseline-band terms are omitted from the table. Random intercepts for patient and lesion slot. The model agrees with the paired tests in direction. Fitted by variational inference; intervals are posterior mean ± 1.96 posterior standard deviations from the variational approximation.

S4. SENSITIVITY ANALYSES

A. Perfusion-Check Threshold

Table S3 gives passes-and-wrong for the topological errors at the 10% threshold and at the 13% and 16% sensitivities.

Table S3. Proportion of Models That Pass the Perfusion Check and Are Materially Wrong, % (Wilson 95% Interval), by Perfusion-Check Threshold

Bed	Errors	Protocol	<i>n</i>	Passes and wrong, % of <i>n</i>			At 10%	
				10%	13%	16%	Pass	Wrong among passing, %
Discrete	T1+T2	A	137	12 (8–19)	23 (17–31)	31 (24–40)	21	81 (60–92)
Discrete	T1+T2	B	137	1 (0–5)	11 (7–17)	20 (14–27)	23	9 (2–27)
Discrete	T1+T2	C	104	19 (13–28)	28 (20–37)	38 (30–48)	39	51 (36–66)
Discrete	T1+T2	D	104	20 (14–29)	20 (14–29)	20 (14–29)	103	20 (14–29)
Leaky	T1+T2	A	218	20 (15–26)	30 (25–37)	38 (31–44)	108	40 (31–49)
Leaky	T1+T2	B	218	5 (3–8)	5 (3–9)	6 (3–9)	176	6 (3–10)
Leaky	T1+T2	C	171	8 (5–13)	9 (6–15)	9 (6–15)	154	9 (5–15)
Leaky	T1+T2	D	171	5 (3–10)	5 (3–10)	5 (3–10)	171	5 (3–10)

Materially wrong: $|\Delta\text{FFR}| > 0.05$. The 13% and 16% thresholds were analyzed as sensitivities. *n* counts models with a defined perfusion residual: T2 under Protocols A and B in the leaky bed, *n* = 100; T2 under Protocol B in the discrete bed, *n* = 60. Protocol D matches every territory flow in all but two models, so its passes-and-wrong and materially wrong proportions differ by at most one model.

B. Hyperemic Demand

With $k = 562 \text{ s}^{-1}$, a vessel with the normal proximal LAD diameter of 3.7 mm carries 214 mL/min, within one standard deviation of the hyperemic flow measured in that artery by continuous thermodilution (228 ± 71 to 293 ± 102 mL/min). The cohort's median demand was 2.28 mL/s (137 mL/min) per tree. The demand constant *k* was scaled by 0.7 and 1.3 with the same cohort (Table S4). The ordering of topological against caliber errors, and of Protocols B and C against Protocol A, was unchanged in both beds; the magnitudes changed because demand moves the clean FFR (leaky-bed median 0.869, 0.801 and 0.741). Tuned topological passes-and-wrong fell to 3% (leaky bed, 0.7 times the demand) and stayed at 12–25% (discrete bed).

Table S4. Sensitivity to Hyperemic Demand

Bed	Demand scale	Flip, % (Protocol A)		Topological flip, %		Passes and wrong, % (C, topological)	Missed-branch ΔFFR median, A / C
		Topological	Caliber	B	C		
Discrete	0.7	32 (25–40)	6 (3–10)	12 (8–18)	18 (12–27)	12 (7–19)	0.070 / 0.057
	1.0	33 (26–41)	10 (7–15)	14 (10–20)	19 (13–28)	19 (13–28)	0.095 / 0.077
	1.3	40 (32–48)	5 (2–9)	16 (11–22)	19 (13–28)	25 (18–34)	0.106 / 0.091
Leaky	0.7	20 (16–26)	6 (4–9)	7 (4–10)	9 (5–14)	3 (1–7)	0.032 / 0.010
	1.0	32 (26–38)	6 (4–9)	5 (3–8)	8 (5–13)	8 (5–13)	0.047 / 0.015
	1.3	37 (32–43)	4 (2–7)	5 (3–9)	8 (4–13)	12 (8–18)	0.057 / 0.021

Demand scale 1.0 is the primary analysis ($k = 562 \text{ s}^{-1}$). Percentages with Wilson 95% intervals in parentheses. Missed-branch ΔFFR is over the instances in which Protocol C was defined.

C. Demand Replication With Re-Selected Cohorts

The whole pipeline was repeated with k doubled and tripled, with cohorts reselected at that demand. At three times the demand only 17 instances met the discrete-bed criteria, so the discrete arm is reported at twice the demand (Table S5). The excess of tuned over re-derived passes-and-wrong at twice the demand was significant in three of four caliber comparisons and in no topological one.

Table S5. Demand Replication: Flip and Passes-and-Wrong Rates, % (Wilson 95% Interval)

Bed	k scale	Inst.	Flip, A		Topological flip		Topological passes and wrong			Caliber passes and wrong		
			Topol.	Caliber	C	D	B	C	D	B	C	D
Discrete	1	97	33 (26–41)	10 (7–15)	19 (13–28)	8 (4–14)	1 (0–5)	19 (13–28)	20 (14–29)	0 (0–2)	5 (2–9)	18 (13–24)
	2	47	50 (38–62)	1 (0–6)	20 (11–32)	8 (3–18)	6 (2–15)	14 (7–26)	20 (11–32)	0 (0–4)	1 (0–6)	14 (8–22)
Leaky	1	150	32 (26–38)	6 (4–9)	8 (5–13)	6 (3–10)	5 (3–8)	8 (5–13)	5 (3–10)	0 (0–1)	12 (9–17)	4 (2–7)
	2	150	34 (29–40)	5 (3–8)	7 (4–11)	3 (1–7)	2 (1–5)	5 (3–10)	3 (1–7)	0 (0–1)	13 (10–18)	5 (3–8)
	3	150	33 (28–39)	5 (3–8)	6 (4–11)	6 (4–11)	4 (2–8)	7 (4–12)	4 (2–8)	0 (0–2)	12 (9–16)	3 (1–5)

Scale 1 is the primary analysis ($k = 562 \text{ s}^{-1}$, primary cohort). Scales 2 and 3 use cohorts reselected at that demand. Passes and wrong: perfusion residual below 10% and $|\Delta\text{FFR}| > 0.05$; denominators are models with a defined residual, as in Table I of the main text.

D. Grey Zone

Flips beyond the grey zone end outside 0.75–0.85 (Table S6); 30% (discrete) and 33% (leaky) of clean models lie within.

Table S6. Flips and Flips Beyond the Grey Zone (0.75–0.85), % of Models (Wilson 95% Interval)

Bed	Protocol	Topological errors			Caliber errors		
		n	Flip	Beyond zone	n	Flip	Beyond zone
Discrete	A	137	33 (26–41)	20 (15–28)	194	10 (7–15)	0 (0–2)
	B	173	14 (10–20)	5 (2–9)	194	8 (5–12)	0 (0–2)
	C	104	19 (13–28)	9 (5–16)	194	8 (5–12)	1 (0–3)
	D	104	8 (4–14)	2 (1–7)	194	11 (7–16)	1 (0–3)
Leaky	A	265	32 (26–38)	23 (18–28)	300	6 (4–9)	0 (0–1)
	B	267	5 (3–8)	0 (0–1)	300	1 (1–3)	0 (0–1)
	C	171	8 (5–13)	1 (0–3)	300	4 (2–7)	1 (0–2)
	D	171	6 (3–10)	0 (0–2)	300	4 (2–6)	0 (0–1)

S5. CALIBER ERROR AT THE STENOSIS THROAT

The throat error (T5) re-inserts each lesion with its throat diameter changed by half the in-plane voxel size (spacing median 0.352 mm; throat radius ± 0.088 mm; 7.3 points of DS, 4.5–9.8) or its DS changed by ± 10 points. The ± 10 -point magnitude lies within the published CT-versus-invasive disagreement (standard deviation 12.3–12.4 points). Protocol B equals Protocol A for this error by construction (Table S7). Protocol C fits at the search bound were excluded (2 at half a voxel, 8 at ± 10 points). The 64 Protocol D fits at the parameter bound, all with an increased stenosis, were retained; excluding them would raise passes-and-wrong by up to 13 points.

Table S7. Throat Caliber Error: Flips, Passes-and-Wrong and Flips Beyond the Grey Zone, % of Models (Wilson 95% Interval)

Error	Protocol	Discrete bed (97 instances)				Leaky bed (150 instances)			
		<i>n</i>	Flip	Passes and wrong	Beyond zone	<i>n</i>	Flip	Passes and wrong	Beyond zone
$\pm \frac{1}{2}$ voxel pooled	A, B	194	30 (24–37)	29 (23–36)	14 (10–20)	300	36 (31–42)	42 (37–48)	21 (16–26)
	C	193	38 (31–45)	57 (50–64)	22 (17–28)	299	37 (32–43)	69 (63–74)	29 (25–35)
	D	194	40 (33–47)	80 (74–85)	26 (21–33)	300	38 (33–44)	81 (76–85)	29 (24–34)
DS ± 10 pooled	A, B	194	40 (33–47)	20 (15–26)	27 (22–34)	300	42 (37–48)	40 (34–45)	35 (30–40)
	C	187	44 (37–52)	51 (44–58)	37 (30–44)	299	43 (38–49)	62 (56–67)	41 (35–46)
	D	194	45 (38–52)	75 (68–80)	40 (34–47)	300	44 (38–49)	86 (81–89)	40 (35–46)
$\pm \frac{1}{2}$ voxel, $k \times 2$	A, B	94	27 (19–36)	35 (26–45)	12 (7–20)	300	29 (24–35)	44 (39–50)	12 (9–16)
	C	94	31 (22–41)	52 (42–62)	13 (7–21)	300	35 (30–40)	69 (64–74)	17 (13–21)
	D	94	34 (25–44)	73 (64–81)	18 (12–27)	300	35 (30–41)	75 (70–80)	18 (14–22)
$\pm \frac{1}{2}$ voxel, $k \times 3$	A, B	–	–	–	–	300	29 (24–34)	48 (43–54)	9 (6–13)
	C	–	–	–	–	300	34 (29–40)	68 (62–73)	15 (12–20)
	D	–	–	–	–	300	33 (28–39)	72 (67–77)	14 (10–18)

$\pm \frac{1}{2}$ voxel: throat diameter narrowed or widened by half the in-plane voxel size, both signs pooled. DS ± 10 : diameter stenosis raised or lowered by 10 points, both signs pooled. Passes and wrong: perfusion residual below 10% and $|\Delta \text{FFR}| > 0.05$. Beyond zone: flips whose corrupted FFR lies outside 0.75–0.85. Protocols A and B coincide for throat errors. Protocol C fits on the search bound are excluded, which reduces *n*. $k \times 2$, $k \times 3$: demand constant doubled or tripled with the cohort reselected at that demand; discrete bed 47 instances at $k \times 2$, not analyzed at $k \times 3$; topological errors under Protocol A flipped 50% (38–62%) and 34% (29–40%) at $k \times 2$ and 33% (28–39%) at $k \times 3$. For comparison, topological errors under Protocol A flipped 33% (26–41%) and 32% (26–38%) of models and caliber errors away from the throat 10% (7–15%) and 6% (4–9%).

S6. TERRITORY GRANULARITY

Protocols C and D were repeated with each main-branch territory divided again at its first bifurcation (median four territories, discrete; five, leaky). A territory without a surviving vessel entered the residual as a 100% mismatch; without this rule, discrete topological passes-and-wrong would have risen to 26%. With the main-branch partition the procedure reproduced the original results exactly (Table S8). Throat-error models removed from the count all failed the finer check; none had its FFR corrected, and 76–88% of tuned throat-error models that passed remained materially wrong. Flip counts changed by at most five per cell. The simulated noise floor was computed at main-branch level only.

Table S8. Passes and Wrong, % (Wilson 95% Interval), at Main-Branch and Finer Territory Level

Bed	Errors	Protocol	n	Main branch	Finer	p
Discrete	T1+T2	C	104	19 (13–28)	7 (3–13)	< 0.001
		D	104	20 (14–29)	3 (1–8)	< 0.001
	T3+T4	C	194	5 (2–9)	5 (3–9)	1.0
		D	194	18 (13–24)	23 (18–30)	0.002
	T5, $\pm \frac{1}{2}$ voxel	A, B	194	29 (23–36)	25 (20–32)	0.016
		C	193	57 (50–64)	51 (44–58)	0.003
Leaky	T1+T2	D	194	80 (74–85)	78 (71–83)	0.12
		C	171	8 (5–13)	5 (2–9)	0.07
	T3+T4	D	171	5 (3–10)	2 (1–6)	0.18
		C	300	12 (9–17)	12 (9–17)	1.0
	T5, $\pm \frac{1}{2}$ voxel	D	300	4 (2–7)	9 (7–13)	< 0.001
		A, B	300	42 (37–48)	35 (30–41)	< 0.001
		C	299	69 (63–74)	60 (55–66)	< 0.001
		D	300	81 (76–85)	81 (76–85)	1.0

Passes and wrong: perfusion residual below 10% and $|\Delta \text{FFR}| > 0.05$. Main branch: subtrees below each child of the first bifurcation (two or three per tree). Finer: each divided again at its own first bifurcation (median four, discrete; five, leaky). *n* counts models with a defined residual at both levels; Protocol C fits on the search bound at either level are excluded. For throat errors the FFR of Protocols A and B is the same at both levels and only the residual changes. *p*: exact McNemar test, paired by model.

S7. OVERLAP AND TOPOLOGY OF THE CORRUPTED TREES

Each tree was represented as cylinders of node radius and element length (a tube model) and compared with its clean tree; no error changed the number of connected components (Table S9). Among the topological errors that changed the decision, 91% (79–96%, discrete) and 62% (51–72%, leaky) kept a whole-scan DSC at or above the inter-observer 0.928. The AUC of tree DSC for a Protocol A flip was 0.65 (discrete) and 0.79 (leaky) across error types and 0.54 and 0.70 within topological errors. The taper DSC exceeds 0.928 because it narrows only the tree beyond the lesion (about 40% of its volume).

Table S9. Tube-Model Overlap of Corrupted Trees, Median (IQR), and Protocol A Decision Changes

Bed	Error	<i>n</i>	DSC, tree	DSC, scan	clDice	Flow lost, %	Flip A, %
Discrete	T1	77	0.975 (0.947–0.984)	0.985 (0.977–0.991)	0.940 (0.903–0.963)	27 (14–43)	27 (19–38)
	T2	96	0.893 (0.701–0.937)	0.935 (0.888–0.960)	0.847 (0.544–0.907)	30 (15–100)	40 (29–53) ^a
	T3	97	0.996 (0.995–0.997)	0.998 (0.998–0.998)	1.000	0	7 (4–14)
	T4	97	0.971 (0.947–0.981)	0.983 (0.974–0.989)	1.000	0	13 (8–22)
Leaky	T1	118	0.969 (0.946–0.980)	0.983 (0.975–0.990)	0.930 (0.899–0.957)	9 (6–17)	18 (12–26)
	T2	149	0.895 (0.721–0.936)	0.934 (0.886–0.962)	0.855 (0.517–0.899)	14 (7–56)	43 (35–51)
	T3	150	0.997 (0.995–0.998)	0.998 (0.998–0.999)	1.000	0	2 (1–6)
	T4	150	0.972 (0.951–0.981)	0.983 (0.976–0.988)	1.000	0	9 (6–15)
T2, right coronary artery			0.62–0.63	0.85–0.86	0.45–0.49		

Flip A: Wilson 95% interval. Flow lost: share of the clean model’s bed outflow on deleted nodes. ^a *n* = 60; right coronary breaks leave no discrete outlet under Protocol A.

S8. DETECTION BEFORE AND DURING TUNING

Positives are error models; negatives are the correct-anatomy draws of the simulated floor before tuning (Table S10). Correct anatomy without noise has a residual below 10^{-8} ; with noise its median is 0.10. With the error models’ targets carrying the same noise, the AUC for topological errors was 0.82 (discrete) and 0.56 (leaky).

Table S10. Detection Before and During Tuning Against Correct Anatomy With Noisy Targets

Bed	Quantity	Positives	<i>n</i>	AUC	Sens. at 0.10, %	FA at 0.10, %	5% FA value	Sens. at 5% FA, %
Discrete	Pre-tuning residual	Topological	137	0.77 (0.71–0.82)	83 (76–89)	48	0.22	47 (39–55)
		Flip under B ^a	36	0.57 (0.46–0.69)	69 (53–82)	48	0.22	22 (12–38)
	ln <i>C</i>	Topological	104	0.47 (0.40–0.53)	30 (22–39)	37	0.24	9 (5–16)
		Flip or P&W under C ^a	61	0.53 (0.44–0.62)	41 (30–54)	37	0.24	15 (8–26)
Leaky	Pre-tuning residual	Topological	218	0.22 (0.18–0.27)	19 (15–25)	50	0.23	6 (3–9)
		Flip under B ^a	16	0.21 (0.08–0.36)	12 (3–36)	50	0.23	6 (1–28)
	ln <i>C</i>	Topological	171	0.12 (0.09–0.15)	3 (1–7)	34	0.24	2 (1–5)
		Flip or P&W under C ^a	66	0.66 (0.55–0.77)	61 (49–71)	34	0.24	59 (47–70)

Negatives: 1 940 (discrete) and 3 000 (leaky; 2 990 for |ln *C*|) correct-anatomy draws on 97 and 150 instances. |ln *C*|: absolute log of the Protocol C scaling applied to the re-derived bed constant. FA, false alarm; P&W, passes and wrong; Sens., sensitivity. ^a Any error type.

S9. NOISE FLOORS

The perfusion-check bound is $1.96 \times (10\text{--}15\%) \approx 20\text{--}29\%$ for one model output against one measurement, from the within-subject variation of hyperemic myocardial blood flow. The simulated floor perturbs, in 20 draws per instance, cardiac output (relative standard deviation 0.05), aortic pressure (0.056), viscosity (0.02), territory flow shares (0.10) and the measured territory flows (0.083), then tunes as in Protocol C (Table S11).

Table S11. Simulated Floor: Correct Anatomy Tuned to Noisy Perfusion

Bed	Instances	Draws solved	Flip, %	Median ΔFFR	95th pct. ΔFFR	Pass, %	Pass and wrong, %
Discrete	97	1940	6.2 (5.2–7.3)	0.012	0.050	72	3.1 (2.4–4.0)
Leaky	150	2990	7.2 (6.3–8.2)	0.012	0.053	72	3.9 (3.2–4.6)

S10. THREE-DIMENSIONAL CASE STUDY: MESH CHECKS

Surfaces were extracted by marching cubes with Taubin smoothing; meshes of 3.5–3.9 million cells (cfMesh) had 25 μm cells within ±4 mm of the throat and four boundary layers. OpenFOAM simpleFoam (ESI v2406; SIMPLE; bounded second-order upwind) ran 3 000 iterations per solve to scaled residuals below 10^{-5} . A finer throat zone (12.5 μm, 6.24 and 8.31 million cells) raised the baseline FFR from 0.8698 to 0.8716 and 0.8718 (0.0019–0.0021) and changed outlet flows by at most 0.25%, above the idealized-case discretization uncertainty of 5.5×10^{-4} , so the patient-case uncertainty is about 0.002, against an error effect of 0.074. The finest mesh missed the strict residual limit (velocity residual 1.08×10^{-5}) with a steady inlet pressure. The lesion carried about 10 mL/min (tree inflow 0.68 mL/s in 3D, 0.65–0.67 mL/s in the cohort model). At the meshed throat radius of 0.276 mm the mean velocity is about 0.7 m/s and the Reynolds number about 110, within the laminar regime.